

SUDEEP K S

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Available: January – August 2027

EDUCATION

Northeastern University, Boston, MA <i>Candidate for a Master of Science in Applied Artificial Intelligence</i> Related courses: Machine Learning, Deep Learning, Natural Language Processing, Applied Generative AI	Expected May 2028 GPA: 3.89/4.0
IIIT Bangalore, Bengaluru, India <i>Advanced Certificate, Data Science (Deep Learning)</i>	Sep 2025 CGPA: 3.89/4.0
BMS Institute of Technology & Management, Bengaluru, India <i>Bachelor of Engineering</i>	2022 CGPA: 7.2/10

TECHNICAL KNOWLEDGE

Languages: Python, SQL
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Hugging Face, OpenCV, FastAPI, Streamlit
Core Skills: Deep Learning, Computer Vision, NLP, LLMs, RAG, Data Visualization (Tableau)
Cloud & DevOps: Azure, Docker, MLflow, Git

WORK EXPERIENCE

Capgemini, Bengaluru, India <i>Senior Software Engineer — RPA & Automation Client: Humana, Inc. (health insurance & wellness services)</i>	Aug 2023 – Dec 2025
- Automated rule-based processes for claims processing, letter generation, and HRA (Health Risk Assessment) workflows using Automation Anywhere 360, reducing production defects by 30% through systematic edge-case coverage. - Partnered with business analysts to gather requirements and author end-to-end RPA solution designs (SDD/ADD) using Visio and Lucidchart across 10+ automated pipelines. - Directed RPA delivery across 5 agile sprint teams using Azure DevOps CI/CD, incorporating unit/integration testing and UAT to reduce deployment errors by 25% and ensure functional and compliance requirements were met. - Wrote SQL queries against Humana's internal data warehouse to retrieve and validate member eligibility data (contract ID, PBP ID, member status) across Medicaid, Medicare, and commercial plans, then confirmed pass/fail execution status via process/transaction ID queries across 10+ automated pipelines. - Monitored production bot performance across 3 platforms, resolved issues through root-cause analysis, maintained 98%+ ServiceNow ticket resolution within SLA, and mentored junior engineers on RPA best practices.	

Rinex Technologies, India <i>Front-End Developer</i>	Sep 2022 – Jul 2023
- Built 8+ responsive forms and UI pages for Rinex's e-learning platform using HTML5, CSS, JavaScript, Bootstrap, and PHP, improving page load performance by 40% through code optimization and lazy loading. - Collaborated closely with a UI/UX designer to translate design mockups into functional, pixel-accurate front-end layouts. - Implemented multi-page navigation flows and PHP-based form handling to support core platform interactions. - Worked within a cross-functional development team to plan, build, and ship features end-to-end on the platform.	

PROJECTS

HealthGuard AI — Multimodal Clinical Intelligence Platform <i>LangGraph / LLMs / Computer Vision / RAG • https://github.com/kssudeep/healthguard-ai</i>	
- Architected a production-grade 5-agent LangGraph system (Vision, NLP, RAG, Critic, Synthesizer) using a supervisor-reflection pattern with circuit breaker logic, generating end-to-end clinical reports in ~12 seconds. - Fine-tuned DenseNet-121 on the NIH ChestX-ray-14 dataset (14 pathologies) with Grad-CAM explainability for interpretable diagnostic predictions. - Implemented BioBERT-based NER for clinical entity extraction with negation detection using SpaCy, improving structured extraction from unstructured clinical text. - Built a hybrid FAISS + BM25 retrieval pipeline with cross-encoder reranking over PubMed and clinical guideline corpora, improving retrieval relevance by 31% over a single-method baseline. - Deployed to Azure Container Apps using Docker, with MLflow experiment tracking (26 metrics/run), Prometheus monitoring, and Redis caching across 5 containerized services.	
OmegaTK RAG Coding Assistant — Commissioned by OpenEye / Cadence Molecular Sciences <i>LangChain / FAISS / RAG / Gemini • https://github.com/kssudeep/omegatkg-rag</i>	
- Built a RAG-based coding assistant for OpenEye's OmegaTK Python toolkit using FAISS + Gemini 2.5 Flash, generating executable molecular conformer code from natural language queries. - Engineered a history-aware, two-step RAG chain (query condensation + MMR retrieval, k=8) over 8 live documentation pages, achieving 15/15 on an automated test suite. - Delivered a Streamlit chat interface with conversation history, inline source citations, and index rebuild controls for end users. - Instrumented a full observability layer with stage-level tracing, p50/p95 latency tracking, and cost-per-request logging via structured JSONL logs, enabling production-grade monitoring of RAG chain performance.	
Pilot Eye Gaze & Hand Gesture Recognition <i>Computer Vision / Deep Learning • https://github.com/kssudeep/Pilot-Eye-Gaze-and-hand-Gesture_Final</i>	
- Built a real-time multimodal computer vision system for pilot eye gaze and hand gesture recognition using OpenCV and MediaPipe. - Fused eye-gaze estimation and hand-landmark tracking into a unified dual-sensor, real-time processing pipeline. - Achieved <50ms inference latency at 30fps, enabling responsive real-time interaction. - Positioned the system for applicability across aviation cockpit monitoring, AR/VR interaction, and accessibility use cases.	